

Homework Set HW4 due 2/16/04 at 12:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt) Section 3.2 problem 12.

$$\text{Let } g(x) = \frac{1}{2\sqrt{x}} + \frac{x^2}{4} + C.$$

$$g'(x) = \underline{\hspace{2cm}}$$

2.(1 pt) Section 3.2 problem 14.

$$\text{Let } g(x) = \frac{2x^5 - 3x^2 + 11}{x^3}.$$

$$g'(x) = \underline{\hspace{2cm}}$$

3.(1 pt) Section 3.2 problem 16.

$$\text{Let } g(x) = (x+2)(2\sqrt{x} + x^2).$$

$$g'(x) = \underline{\hspace{2cm}}$$

4.(1 pt) Section 3.2 problem 18.

$$\text{Let } f(x) = \frac{(x^2 + 3)}{(x^2 + 5)}.$$

$$f'(x) = \underline{\hspace{2cm}}$$

5.(1 pt) Section 3.2 problem 21.

$$\text{Let } f(x) = x^5 - 5x^3 + x + 12. \text{ Find}$$

$$f'(x) = \underline{\hspace{2cm}} \quad f''(x) = \underline{\hspace{2cm}} \quad f'''(x) = \underline{\hspace{2cm}}$$

$$f^{(4)}(x) = \underline{\hspace{2cm}}$$

6.(1 pt) This is problem 27, Section 3.2 page 118.

$$\text{If } f(x) = x^2 - 3x - 5, \text{ then } f'(-2) = \underline{\hspace{2cm}}$$

Use this to find the equation of the tangent line to the parabola $y = x^2 - 3x - 5$ when $x = -2$.

The equation of this tangent line can be written in the form $y = mx + b$ where m is: $\underline{\hspace{2cm}}$

and where b is: $\underline{\hspace{2cm}}$

7.(1 pt) This is problem 28, Section 3.2 page 118.

$$\text{If } f(x) = x^5 - 3x^3 - 5x + 2, \text{ find } f'(1). \underline{\hspace{2cm}}$$

Use this to find the equation of the tangent line to the graph of $y = x^5 - 3x^3 - 5x + 2$ when $x = 1$.

The equation of this tangent line can be written in the form $y = mx + b$ where m is: $\underline{\hspace{2cm}}$

and where b is: $\underline{\hspace{2cm}}$

8.(1 pt) This is problem 30, Section 3.2 page 118.

$$\text{If } f(x) = \frac{(x+1)}{(x-1)}, \text{ find } f'(0). \underline{\hspace{2cm}}$$

Use this to find the equation of the tangent line to the graph of $y = \frac{(x+1)}{(x-1)}$ when $x = 0$.

The equation of this tangent line can be written in the form $y = mx + b$ where m is: $\underline{\hspace{2cm}}$

and where b is: $\underline{\hspace{2cm}}$

9.(1 pt) This is problem 32, Section 3.2 page 118.

$$\text{If } f(x) = 1 - \frac{1}{x} + \frac{2}{\sqrt{x}}, \text{ find } f'(4). \underline{\hspace{2cm}}$$

Use this to find the equation of the tangent line to the graph of $y = 1 - \frac{1}{x} + \frac{2}{\sqrt{x}}$ when $x = 4$.

The equation of this tangent line can be written in the form $y = mx + b$ where m is: $\underline{\hspace{2cm}}$

and where b is: $\underline{\hspace{2cm}}$

10.(1 pt) This is problem 34, Section 3.2 page 118.

If $f(t) = t^4 + 4t^3 - 8t^2 + 3$, then there are (one or more) points with coordinates $(t, f(t))$ at which $f(t)$ has a horizontal tangent line.

Find all such points. Then give the sum of the t coordinates of these points $\underline{\hspace{2cm}}$

11.(1 pt) This is problem 36, Section 3.2 page 118.

If $f(t) = \frac{1}{t^2} - \frac{1}{t^3}$, then there are (one or more) points with coordinates $(t, f(t))$ at which $f(t)$ has a horizontal tangent line.

Find all such points. Then give the sum of the t coordinates $\underline{\hspace{2cm}}$